COS 568 Assignment 2: Distributed Training of Language Models

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1 Setup and Reproducibility

- Task 1 CPU run: 1 node (node202), backend `gloo`-free single-process training.
- Task 1 GPU run: 1 node (node915), single GPU.
- CPU distributed timing runs: Tasks 2(a), 2(b), and 3 on `node202, node203, node204, node206`; 1 rank per node, backend `gloo`.
- CPU Task 4 profiling runs: `node701, node702, node703, node704`; 1 rank per node, backend `gloo`.
- Multi-node GPU distributed runs: 4 nodes, 4 ranks, 1 rank per node, backend `nccl`.
- GPU distributed nodes: `node105, node915, node916, node917`
- Seed: `42`, PyTorch version: `2.9.1+cu128`, commit: `4444ce4`.

2 Task 1: Single-Node Fine-Tuning

Epoch	CPU acc	GPU acc
1	0.5920577617	0.5740072202
2	0.6462093863	0.6389891697
3	0.6534296029	0.6606498195

Table 1: Task 1 evaluation accuracy after each epoch.

Minibatch	CPU loss	GPU loss
1	0.748251	0.778352
2	0.797213	0.785116
3	0.708992	0.688311
4	0.755476	0.738948
5	0.740575	0.708527

Table 2: Task 1 first five minibatch losses.

3 Tasks 2(a), 2(b), and 3

Hardware	Task	Method	Avg iter time (s)	Timed iters
CPU	2(a)	gather+scatter	5.513219	38
CPU	2(b)	all_reduce	3.957924	38
CPU	3	DDP	3.322300	38
GPU	2(a)	gather+scatter	2.335618	38
GPU	2(b)	all_reduce	0.930466	38
GPU	3	DDP	0.680639	38

Table 3: Average iteration time after dropping the first iteration.

CPU Task 2(a) vs 2(b) max per-step loss difference: 1.788e-07.
GPU Task 2(a) vs 2(b) max per-step loss difference: 1.788e-07.

CPU Task 2(a) **GPU Task 2(a)**

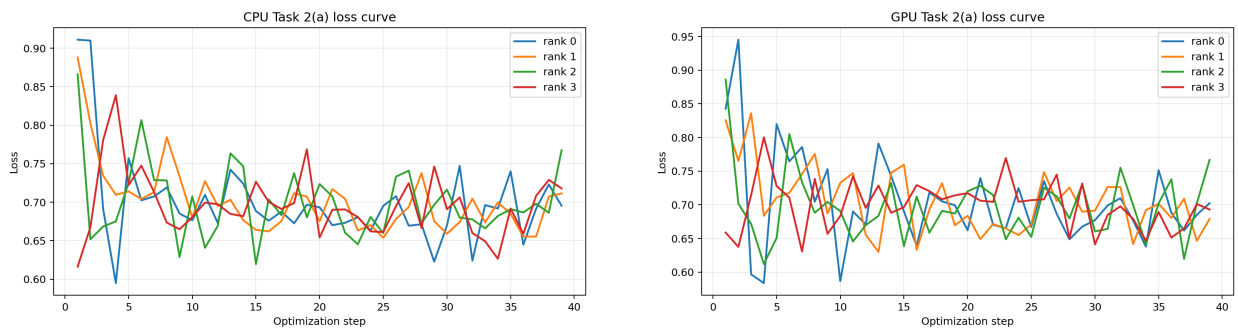

Figure 1: Task 2(a) loss curves across all ranks.

CPU Task 2(b) **GPU Task 2(b)**

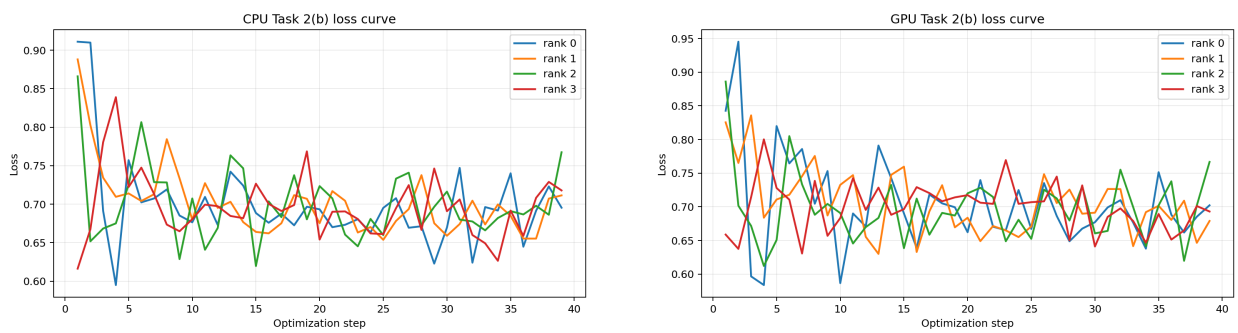

Figure 2: Task 2(b) loss curves across all ranks.

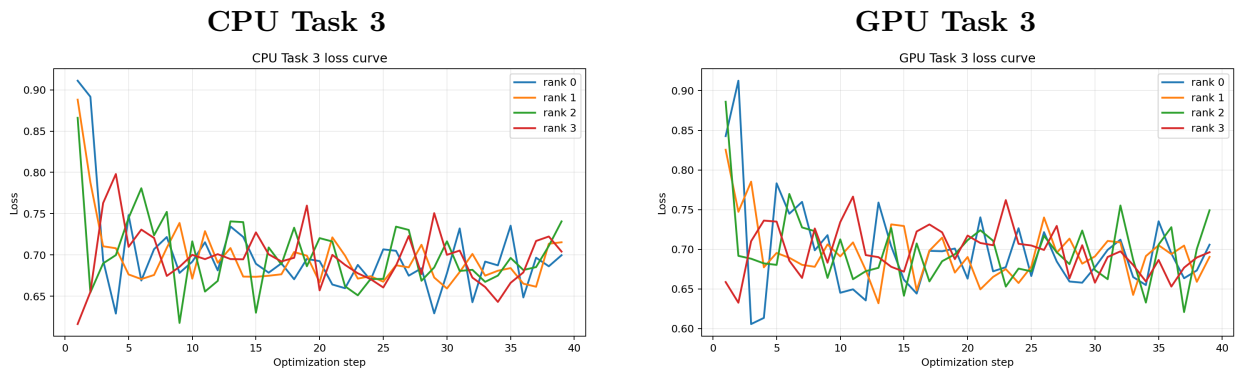

Figure 3: Task 3 loss curves across all ranks.

4 Task 4: Profiling and Communication Overhead

Hardware	Method	Step	Total ms	Comm ms / pct
CPU	gather_scatter	1	27025.274	21229.106 / 78.553%
CPU	gather_scatter	2	27012.713	21243.450 / 78.642%
CPU	gather_scatter	3	27019.735	21237.381 / 78.600%
CPU	all_reduce	1	11033.723	5551.403 / 50.313%
CPU	all_reduce	2	11031.779	5553.455 / 50.341%
CPU	all_reduce	3	11086.272	5557.129 / 50.126%
CPU	ddp	1	8296.159	5298.015 / 63.861%
CPU	ddp	2	8123.591	5267.552 / 64.843%
CPU	ddp	3	8153.338	5276.082 / 64.711%
GPU	gather_scatter	1	2874.860	2139.384 / 74.417%
GPU	gather_scatter	2	2332.101	2174.620 / 93.247%
GPU	gather_scatter	3	2325.256	2168.194 / 93.245%
GPU	all_reduce	1	859.100	716.353 / 83.384%
GPU	all_reduce	2	895.525	754.992 / 84.307%
GPU	all_reduce	3	897.617	760.135 / 84.684%
GPU	ddp	1	728.845	653.159 / 89.616%
GPU	ddp	2	717.997	647.054 / 90.119%
GPU	ddp	3	707.342	635.475 / 89.840%

Table 4: Per-step communication overhead from the profiled three-step windows.

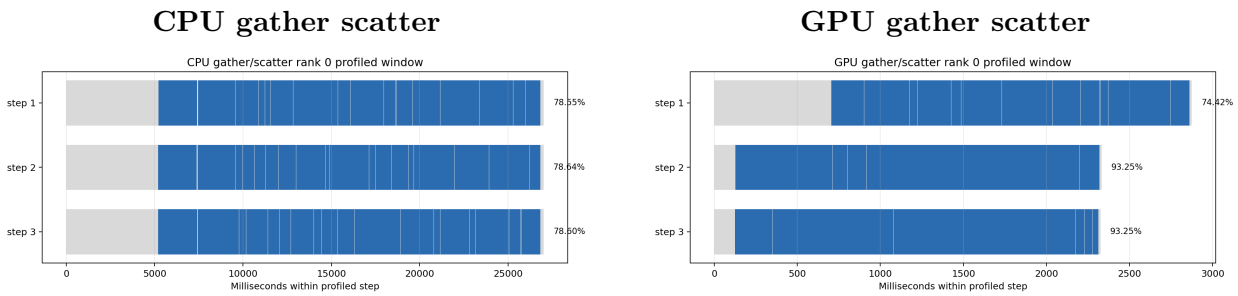

Figure 4: Profiler overview for gather scatter over profiled steps 1-3.

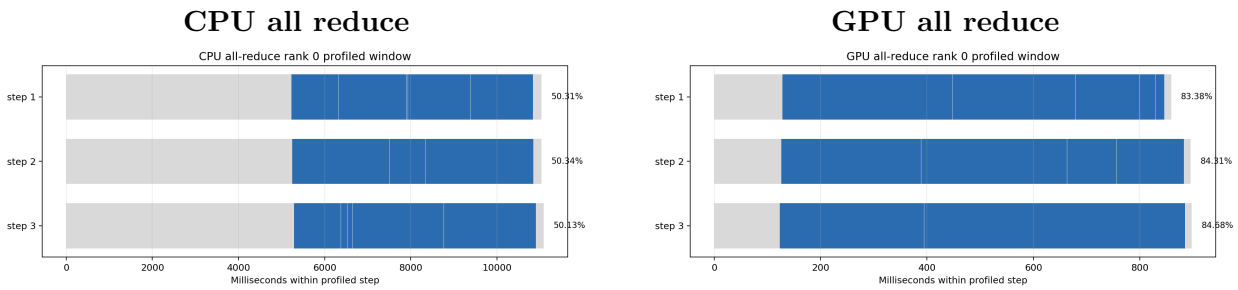

Figure 5: Profiler overview for all reduce over profiled steps 1-3.

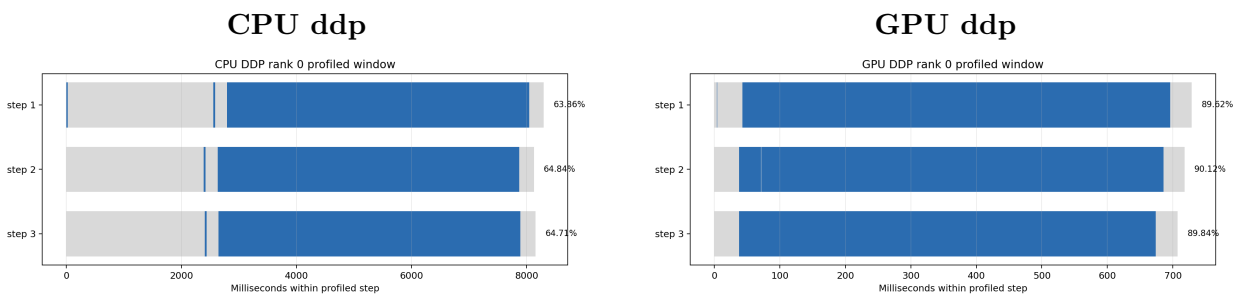

Figure 6: Profiler overview for ddp over profiled steps 1-3.

5 Discussion

The old CPU numbers in this repo came from a one-node debug launcher. The rerun above uses true multi-node CPU and GPU jobs with 1 rank per node. GPU training is still faster end-to-end: the new GPU speedups over CPU are 2.36x for Task 2(a), 4.25x for Task 2(b), and 4.88x for Task 3.

CPU Task 2(b) is 28.21% faster than CPU Task 2(a), and CPU DDP is another 16.06% faster than CPU all-reduce on the same 4-node allocation.

For Task 4, CPU DDP has an average communication-overhead percentage of 64.471% versus 50.260% for manual all-reduce, a relative reduction of -28.276%. On GPU, the same trace-derived metric is 89.858% for DDP versus 84.125% for all-reduce, so the relative reduction is -6.815%. Even when the GPU trace metric is not lower for DDP, DDP still wins on end-to-end iteration time.